
Polynomials: Advanced

Jacob Khohayting

September 20, 2026

CONTENTS

1	Lagrange Interpolation	2
2	Newton Interpolation	2
3	Newton Finite Differences	2
4	Selected Problems	3
5	Solutions	4
5.1	2024 MPfG	4
5.2	2021 CMIMC	5
5.3	1975 USAMO	6
5.4	2025 MPfG	7
5.5	2022 BMT	8
5.6	2024 HMMT	9
5.7	2025 FLSAM	10
5.8	2019 Putnam	11

1 LAGRANGE INTERPOLATION

Let P be a polynomial passing through the points (x_i, y_i) for $0 \leq i \leq n$. The *unique* polynomial of minimal degree is

$$P(x) = \sum_{i=0}^n \left(y_i \cdot \frac{\prod_{j \neq i} (x - x_j)}{\prod_{j \neq i} (x_i - x_j)} \right)$$

We will see for $n = 2$. Suppose $P(x)$ is the polynomial of minimal degree passing through (x_1, y_1) and (x_2, y_2) . This is evidently a linear polynomial which we can solve geometrically. Firstly, the slope of the polynomial passing through these points is $\frac{y_2 - y_1}{x_2 - x_1}$. Now:

$$\begin{aligned} P(x) &= \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1) + y_1 \\ &= \frac{y_2}{x_2 - x_1} \cdot (x - x_1) + \frac{y_1}{x_1 - x_2} \cdot (x - x_1) + y_1 \\ &= \frac{y_2}{x_2 - x_1} \cdot (x - x_1) + \frac{y_1}{x_1 - x_2} \cdot (x - x_2). \end{aligned}$$

This clearly matches the form as shown above.

2 NEWTON INTERPOLATION

This follows via rearranging the above. Define $[y_0] = y_0$ and

$$[y_0, y_1, \dots, y_k] = \frac{[y_1, y_2, \dots, y_k] - [y_0, y_1, \dots, y_{k-1}]}{x_k - x_0}.$$

Let P be a polynomial passing through the points (x_i, y_i) for $0 \leq i \leq n$. Then:

$$P(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)(x - x_1) + \dots + c_n(x - x_0)(x - x_1) \dots (x - x_{n-1}).$$

Then, $c_n = [y_0, y_1, \dots, y_n]$.

3 NEWTON FINITE DIFFERENCES

Suppose P is a polynomial, and suppose $c_0 = P(1) - P(0)$, $c_1 = P(2) - 2P(1) + P(0)$, $c_2 = P(3) - 3P(2) + 3P(1) - P(0)$, etc. Then:

$$P(x) = \sum_{n=0}^{\infty} c_n \binom{x}{n}.$$

4 SELECTED PROBLEMS

Problem 1. (2024 MPfG) Let p be the unique polynomial of degree 6 such that

$$p(n) = (-1)^n \binom{6}{n}$$

for $n = 0, 1, 2, 3, \dots, 6$. What is $p(7)$?

Problem 2. (2021 CMIMC) Suppose f is a degree 42 polynomial such that for all integers $0 \leq i \leq 42$,

$$f(i) + f(43+i) + f(2 \cdot 43+i) + \dots + f(46 \cdot 43+i) = (-2)^i.$$

Find $f(2021) - f(0)$.

Problem 3. (1975 USAMO) If $P(x)$ denotes a polynomial of degree n such that $P(k) = \frac{k}{k+1}$ for $k = 0, 1, 2, \dots, n$, determine $P(n+1)$.

Problem 4. (2025 MPfG) Solve for a :

$$\begin{aligned} a - b + c - d + e &= 1 \\ a - 2b + 4c - 8d + 16e &= 128 \\ a - 3b + 9c - 27d + 81e &= 2187 \\ a - 4b + 16c - 64d + 256e &= 16384 \\ a - 5b + 25c - 125d + 625e &= 78125. \end{aligned}$$

Problem 5. (2022 BMT) Let $x_1, x_2, \dots, x_{2022}$ be complex numbers such that

$$\begin{aligned} x_1 x_2 \cdots x_{2022} &= 1, \\ (x_1 + 1)(x_2 + 1) \cdots (x_{2022} + 1) &= 2, \\ &\vdots \\ (x_1 + 2021)(x_2 + 2021) \cdots (x_{2022} + 2021) &= 2^{2021}. \end{aligned}$$

Find $(x_1 + 2022)(x_2 + 2022) \cdots (x_{2022} + 2022)$.

Problem 6. (2024 HMMT) Let $f(x)$ be a quotient of two quadratic polynomials. Given that $f(n) = n^3$ for all $n \in \{1, 2, 3, 4, 5\}$, compute $f(0)$.

Problem 7. (2025 FLSAM) For each nonnegative integer k , $P_k(x)$ is a real polynomial of degree k with leading coefficient $\frac{k+1}{k!}$. For each k , we have that

$$P_k(k) = P_{k+1}(k) = P_{k+2}(k) = \dots$$

Given that $P_{100}(100)$ is an integer, compute the number of positive integer divisors of $P_{100}(100)$.

Problem 8. (2019 Putnam) Let F_m be the m -th Fibonacci number, defined by $F_1 = F_2 = 1$ and $F_m = F_{m-1} + F_{m-2}$ for all $m \geq 3$. Let $p(x)$ be the polynomial of degree 1008 such that $p(2n+1) = F_{2n+1}$ for $n = 0, 1, 2, \dots, 1008$. Find integers j and k such that $p(2019) = F_j - F_k$.

5 SOLUTIONS

5.1 2024 MPFG

Let p be the unique polynomial of degree 6 such that

$$p(n) = (-1)^n \binom{6}{n}$$

for $n = 0, 1, 2, 3, \dots, 6$. What is $p(7)$?

Solution. By Lagrange Interpolation:

$$\begin{aligned} p(7) &= \sum_{i=0}^6 \left((-1)^i \binom{6}{i} \cdot \frac{\prod_{j \neq i} (7-j)}{\prod_{j \neq i} (i-j)} \right) \\ &= \sum_{i=0}^6 \left((-1)^i \binom{6}{i} \cdot \frac{7!/(7-i)}{i! \cdot (6-i)! \cdot (-1)^{6-i}} \right) \\ &= \sum_{i=0}^6 \left(\binom{6}{i} \cdot \frac{7!}{i! \cdot (7-i)!} \right) \\ &= \sum_{i=0}^6 \left(\binom{6}{i} \cdot \binom{7}{7-i} \right) \\ &= \binom{13}{7} \\ &= \boxed{1716}. \end{aligned}$$

5.2 2021 CMIMC

Suppose f is a degree 42 polynomial such that for all integers $0 \leq i \leq 42$,

$$f(i) + f(43 + i) + f(2 \cdot 43 + i) + \cdots + f(46 \cdot 43 + i) = (-2)^i.$$

Find $f(2021) - f(0)$.

Solution. Let $P(i) = f(i) + f(43 + i) + f(2 \cdot 43 + i) + \cdots + f(46 \cdot 43 + i)$. Then, $P(i) = (-2)^i$ for all $0 \leq i \leq 42$. Notice,

$$P(43) - P(0) = f(2021) - f(0).$$

By Lagrange Interpolation,

$$\begin{aligned} P(43) &= \sum_{i=0}^{42} (-2)^i \cdot \frac{\prod_{j \neq i} (43 - j)}{\prod_{j \neq i} (i - j)} \\ &= \sum_{i=0}^{42} (-2)^i \cdot \frac{43! / (43 - i)}{i! \cdot (42 - i)! \cdot (-1)^{42-i}} \\ &= \sum_{i=0}^{42} 2^i \cdot \frac{43!}{i! \cdot (43 - i)!} \\ &= \sum_{i=0}^{42} 2^i \cdot \binom{43}{i} \\ &= \left(\sum_{i=0}^{43} 2^i \cdot \binom{43}{i} \right) - 2^{43} \cdot \binom{43}{43} \\ &= 3^{43} - 2^{43}. \end{aligned}$$

But, $P(0) = 1$. Thus, $f(2021) - f(0) = \boxed{3^{43} - 2^{43} - 1}.$

5.3 1975 USAMO

If $P(x)$ denotes a polynomial of degree n such that $P(k) = \frac{k}{k+1}$ for $k = 0, 1, 2, \dots, n$, determine $P(n+1)$.

Solution. Consider the polynomial $P(k) \cdot (k+1) - k = 0$ with roots $0, 1, \dots, n$. Therefore:

$$P(k) \cdot (k+1) - k = C \cdot k(k-1)(k-2) \cdots (k-n),$$

for some constant C . Then, when $k = -1$:

$$1 = C \cdot (-1)(-2) \cdots (-1-n) = C \cdot (-1)^{n+1} \cdot (n+1)! \implies C = \frac{1}{(-1)^{n+1} \cdot (n+1)!}.$$

Furthermore, when $k = n+1$:

$$P(n+1) \cdot (n+2) - (n+1) = C \cdot (n+1)! = \frac{(n+1)!}{(-1)^{n+1} \cdot (n+1)!} = (-1)^{n+1}.$$

Rearranging, then $P(n+1) = \boxed{\frac{(n+1) + (-1)^{n+1}}{n+2}}.$

5.4 2025 MPFG

Solve for a :

$$\begin{aligned} a - b + c - d + e &= 1 \\ a - 2b + 4c - 8d + 16e &= 128 \\ a - 3b + 9c - 27d + 81e &= 2187 \\ a - 4b + 16c - 64d + 256e &= 16384 \\ a - 5b + 25c - 125d + 625e &= 78125. \end{aligned}$$

Solution. We will wrap in a polynomial. For $x \in \{-1, -2, -3, -4, -5\}$, then

$$x^7 + ex^4 + dx^3 + cx^2 + bx + a = (x+1)(x+2)(x+3)(x+4)(x+5)(x^2 + mx + n).$$

Crucially, the coefficients of x^6 and x^5 are 0. So, we will first compute the x^4 and x^3 coefficients of

$$(x+1)(x+2)(x+3)(x+4)(x+5).$$

- The x^4 coefficient is the negated sum of the roots: $1 + 2 + 3 + 4 + 5 = 15$.

- The x^3 coefficient is

$$\frac{(1+2+\cdots+5)^2 - (1^2 + 2^2 + \cdots + 5^2)}{2} = \frac{15^2 - 55}{2} = 85.$$

Therefore, the coefficients of x^6 and x^5 are:

$$m + 15 = 0 \implies m = -15.$$

$$85 + (-15) \cdot 15 + n = 0 \implies n = 140.$$

Thus, $a = 5! \cdot 140 = \boxed{16800}$.

Remark. Try to find a solution with *Lagrange Interpolation*. You know the values of $p(x)$ for $x \in \{-1, -2, -3, -4, -5\}$.

5.5 2022 BMT

Let $x_1, x_2, \dots, x_{2022}$ be complex numbers such that

$$\begin{aligned} x_1 x_2 \cdots x_{2022} &= 1, \\ (x_1 + 1)(x_2 + 1) \cdots (x_{2022} + 1) &= 2, \\ &\vdots \\ (x_1 + 2021)(x_2 + 2021) \cdots (x_{2022} + 2021) &= 2^{2021}. \end{aligned}$$

Find $(x_1 + 2022)(x_2 + 2022) \cdots (x_{2022} + 2022)$.

Solution. Let $P(N) = (N + x_1)(N + x_2) \cdots (N + x_{2022})$. Then, for $N \in \{0, 1, \dots, 2021\}$, we require $P(N) = 2^N$. By Lagrange Interpolation, consider the polynomial:

$$\begin{aligned} Q(2022) &= \sum_{i=0}^{2021} \left(2^i \cdot \frac{\prod_{j \neq i} (2022 - j)}{\prod_{j \neq i} (i - j)} \right) \\ &= \sum_{i=0}^{2021} \left(2^i \cdot \frac{2022!}{(2022 - i) \cdot i! \cdot (2021 - i)! \cdot (-1)^{2021-i}} \right) \\ &= \sum_{i=0}^{2021} (-1) \cdot (-2)^i \binom{2022}{i} \\ &= - \left(\sum_{i=0}^{2022} (-2)^i \binom{2022}{i} \right) + ((-2)^{2022}) \\ &= -((-1)^{2022}) + ((-2)^{2022}) \\ &= 2^{2022} - 1. \end{aligned}$$

Now, $P(N) - Q(N) = 0$ for $\{0, 1, \dots, 2021\}$. This is also a monic polynomial with degree 2022. Hence,

$$P(N) - Q(N) = N(N - 1) \cdots (N - 2021).$$

Thus, $P(2022) - Q(2022) = 2022!$. Hence, $P(2022) = \boxed{2022! + 2^{2022} - 1}$.

5.6 2024 HMMT

Let $f(x)$ be a quotient of two quadratic polynomials. Given that $f(n) = n^3$ for all $n \in \{1, 2, 3, 4, 5\}$, compute $f(0)$.

Solution. Suppose $f(n) = \frac{g(n)}{h(n)}$. Then:

$$f(n) = \frac{g(n)}{h(n)} = n^3 \implies h(n) \cdot n^3 - g(n) = 0,$$

for $n \in \{1, 2, 3, 4, 5\}$. Thus, for a constant C ,

$$\begin{aligned} h(n) \cdot n^3 - g(n) &= C \cdot (n-1)(n-2)(n-3)(n-4)(n-5) \\ &= C \cdot (n^5 - 15n^4 + 85n^3 - 225n^2 + 274n - 120) \\ &= (C \cdot (n^2 - 15n + 85)) \cdot n^3 - (C \cdot (225n^2 - 274n + 120)). \end{aligned}$$

Thus, $h(n) = n^2 - 15n + 85$ and $g(n) = 225n^2 - 274n + 120$. Therefore, $h(0) = 85$ and $g(0) = 120$. Thus,

$$f(0) = \frac{g(0)}{h(0)} = \frac{120}{85} = \boxed{\frac{24}{17}}.$$

5.7 2025 FLSAM

For each nonnegative integer k , $P_k(x)$ is a real polynomial of degree k with leading coefficient $\frac{k+1}{k!}$. For each k , we have that

$$P_k(k) = P_{k+1}(k) = P_{k+2}(k) = \cdots$$

Given that $P_{100}(100)$ is an integer, compute the number of positive integer divisors of $P_{100}(100)$.

Solution. Let $c_k = k + 1$, so that $P_k(x)$ has leading coefficient $\frac{c_k}{k!}$. Let $a_k = P_k(k)$. Then we are given that for each k ,

$$(a_0, a_1, \dots, a_k) = (P_k(0), P_k(1), \dots, P_k(k)).$$

Note that $P_k(x) - P_{k-1}(x)$ has roots at $0, 1, \dots, k-1$. This implies that

$$P_k(x) = P_{k-1}(x) + \frac{c_k}{k!} x(x-1) \cdots (x-k+1) = P_{k-1}(x) + c_k \binom{x}{k}.$$

Thus,

$$P_k(x) = c_k \binom{x}{k} + c_{k-1} \binom{x}{k-1} + \cdots + c_0 \binom{x}{0}.$$

We now compute:

$$\begin{aligned} P_{100}(100) &= 101 \binom{100}{100} + 100 \binom{100}{99} + \cdots + 1 \binom{100}{0} \\ &= \left(\binom{100}{100} + \binom{100}{99} + \cdots + \binom{100}{0} \right) + 100 \left(\binom{99}{99} + \binom{99}{98} + \cdots + \binom{99}{0} \right) \\ &= 2^{100} + 100 \cdot (2^{99}) \\ &= 2^{100} \cdot 3 \cdot 17. \end{aligned}$$

Then the desired value is 404.

5.8 2019 PUTNAM

Let F_m be the m -th Fibonacci number, defined by $F_1 = F_2 = 1$ and $F_m = F_{m-1} + F_{m-2}$ for all $m \geq 3$. Let $p(x)$ be the polynomial of degree 1008 such that $p(2n+1) = F_{2n+1}$ for $n = 0, 1, 2, \dots, 1008$. Find integers j and k such that $p(2019) = F_j - F_k$.

Solution. By Lagrange Interpolation:

$$\begin{aligned}
 p(2019) &= \sum_{i=0}^{1008} F_{2i+1} \cdot \frac{\prod_{j \neq i} (2019 - (2j+1))}{\prod_{j \neq i} ((2i+1) - (2j+1))} = \sum_{i=0}^{1008} F_{2i+1} \cdot \frac{\prod_{j \neq i} (2018 - 2j)}{\prod_{j \neq i} (2i - 2j)} \\
 &= \sum_{i=0}^{1008} F_{2i+1} \cdot \frac{\prod_{j \neq i} (1009 - j)}{\prod_{j \neq i} (i - j)} \\
 &= \sum_{i=0}^{1008} F_{2i+1} \cdot \frac{1009!}{(1009 - i) \cdot \prod_{j \neq i} (i - j)} \\
 &= \sum_{i=0}^{1008} F_{2i+1} \cdot \frac{1009!}{i! \cdot (1009 - i)! \cdot (-1)^{1009-i}} \\
 &= \sum_{i=0}^{1008} F_{2i+1} \cdot \binom{1009}{i} \cdot (-1)^{i+1}
 \end{aligned}$$

By Binet's Formula, $F_i = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^i - \left(\frac{1-\sqrt{5}}{2} \right)^i \right)$. Therefore,

$$\begin{aligned}
 p(2019) &= \sum_{i=0}^{1008} F_{2i+1} \cdot \binom{1009}{i} \cdot (-1)^{i+1} \\
 &= \sum_{i=0}^{1008} \left(\frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^{2i+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{2i+1} \right) \right) \cdot \binom{1009}{i} \cdot (-1)^{i+1} \\
 &= \frac{1}{\sqrt{5}} \cdot \sum_{i=0}^{1008} \left(\left(\frac{1+\sqrt{5}}{2} \right)^{2i+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{2i+1} \right) \cdot \binom{1009}{i} \cdot (-1)^{i+1} \\
 &= \frac{1}{\sqrt{5}} \cdot \sum_{i=0}^{1008} \left(\left(\frac{1+\sqrt{5}}{2} \right)^{2i+1} \cdot \binom{1009}{i} \cdot (-1)^{i+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{2i+1} \cdot \binom{1009}{i} \cdot (-1)^{i+1} \right) \\
 &= -\frac{1+\sqrt{5}}{2\sqrt{5}} \cdot \sum_{i=0}^{1008} \left(\left(\frac{3+\sqrt{5}}{2} \right)^i \cdot \binom{1009}{i} \cdot (-1)^i \right) + \frac{1-\sqrt{5}}{2\sqrt{5}} \cdot \sum_{i=0}^{1008} \left(\left(\frac{3-\sqrt{5}}{2} \right)^i \cdot \binom{1009}{i} \cdot (-1)^i \right) \\
 &= -\frac{1+\sqrt{5}}{2\sqrt{5}} \cdot \left(\left(\frac{1+\sqrt{5}}{2} \right)^{1009} - \left(\frac{3+\sqrt{5}}{2} \right)^{1009} \right) + \frac{1-\sqrt{5}}{2\sqrt{5}} \cdot \left(\left(\frac{1-\sqrt{5}}{2} \right)^{1009} - \left(\frac{3-\sqrt{5}}{2} \right)^{1009} \right) \\
 &= -\frac{1+\sqrt{5}}{2\sqrt{5}} \cdot \left(\left(\frac{1+\sqrt{5}}{2} \right)^{1009} - \left(\frac{1+\sqrt{5}}{2} \right)^{2018} \right) + \frac{1-\sqrt{5}}{2\sqrt{5}} \cdot \left(\left(\frac{1-\sqrt{5}}{2} \right)^{1009} - \left(\frac{1-\sqrt{5}}{2} \right)^{2018} \right) \\
 &= -\frac{1}{\sqrt{5}} \cdot \left(\left(\frac{1+\sqrt{5}}{2} \right)^{1010} - \left(\frac{1+\sqrt{5}}{2} \right)^{2019} \right) + \frac{1}{\sqrt{5}} \cdot \left(\left(\frac{1-\sqrt{5}}{2} \right)^{1010} - \left(\frac{1-\sqrt{5}}{2} \right)^{2019} \right) \\
 &= \frac{1}{\sqrt{5}} \cdot \left(\left(\frac{1+\sqrt{5}}{2} \right)^{2019} - \left(\frac{1-\sqrt{5}}{2} \right)^{2019} \right) - \frac{1}{\sqrt{5}} \cdot \left(\left(\frac{1+\sqrt{5}}{2} \right)^{1010} - \left(\frac{1-\sqrt{5}}{2} \right)^{1010} \right) \\
 &= F_{2019} - F_{1010}.
 \end{aligned}$$

Thus, $(j, k) = \boxed{(2019, 1010)}$.